

S — Set (session)		
Short	Long	Description
SS	SetSpeed	Set cruise speed mm/s ($\leq$ max_speed); bare reloads init_speed; applies live to the next fill.
SA	SetAccel	Set accel mm/s ² ($\leq$ max_accel); bare reloads init_accel; applies live to the next fill.
SE	SetEnable	Driver enable 0 1; bare toggles; required before motion; off stops hard.
ST	SetTerminal	Terminal Mode 0 1; bare toggles; local echo + UART command sniff to USB.
SV	SetVerbose	Verbose status push 0 1; bare toggles; ~3 Hz #M/#I lines when on.
SD	SetDebug	USB-only debug level 0..5; bare restores default; never sent on UIC UART.

G — Get (session)		
Short	Long	Description
GS	GetSpeed	Reply GS:<mm/s> — current session cruise speed.
GA	GetAccel	Reply GA:<mm/s2> — current session acceleration.
GE	GetEnable	Reply GE:0 1 — driver enable state.
GT	GetTerminal	Reply GT:0 1 — Terminal Mode state.
GV	GetVerbose	Reply GV:0 1 — verbose push state.
GD	GetDebug	Reply GD:<0..5> — USB debug level.

I — Is / status		
Short	Long	Description
IM	IsMoving	Reply IM:0 1 — axis currently moving (or settling).
IH	IsHoming	Reply IH:0 1 — homing cycle active.
IL	IsLimit	Reply IL:0 1 — at soft-limit position.
IE	IsError	Reply IE:0 1 — PIN_DRV_ERROR / EMO latched.
IP	IsPosition	Reply IP:<mm> or IP:<mm1> <mm2> when axis2 on.
IA	IsAxis	Reply IA:1 2 — active axis count.
IT	IsTarget	Reply IT:<mm> - — seek target, or - if none.
IR	IsReady	Reply IR:1 only if idle, not homing, enabled, and not waiting.
IW	IsWaiting	Reply IW:1 if any W / WM / WH wait is active.
ID	IsDiag	Reply underrun count, peak STEP Hz, overshoot steps, min FIFO level.
IZ	IsReset	Reply last chip reset cause (power wdt run soft debug brownout ...).
IX	Pinout	ASCII table: GP number, pin name, brief description ($\leq$ 80 cols).

M — Movement		
Short	Long	Description
MT	MoveTo	Absolute mm [optional 2nd axis]; skip none/N/_/*; live-retarget.
M	Move	Relative mm [optional 2nd axis]; alias MoveBy; live-retarget.
ML	MoveLeft	Jog -; optional mask 0 1 2 when axis2; soft-stop MS/!.
MR	MoveRight	Jog +; optional mask 0 1 2 when axis2; soft-stop MS/!.
MH	MoveHome	Homing; optional axis 1 2 when axis2; cancel MS/H.
MS	MoveStop	Soft decelerate to stop; keeps enable; does not cancel waits.

X — Extender		
Short	Long	Description
X0~3	Ext0~3	Ext out n logical 0 1; bare toggles; glued X00≡X0 0; ok during EMO.

C — Config		
Short	Long	Description
CS	ConfigSet	Set persistent key value (mc.ini); silent ok; updates session init_speed/init_accel/...
CR	ConfigReset	Reset all config to compiled defaults and save mc.ini.
CG	ConfigGet	Get key → CG:key=value; bare dumps all keys.
RB	Reboot	Soft MCU reset (no power cycle); EN off first. Use after CS axis2_use.

W — Wait		
Short	Long	Description
W	Wait	Delay sec then continue ; chain; bare → 1 s; never !E:timeout.
WM	WaitMoving	Pause chain until move ends; optional timeout cancels remaining chain.
WH	WaitHoming	Pause chain until homing ends; optional timeout cancels remaining chain.

V — Version		
Short	Long	Description
VA	VersionAbout	Reply VA: about string (name, version, author).
VF	VersionFW	Reply VF:<version> — firmware version.
VP	VersionProtocol	Reply VP:<n> — protocol version.

Special		
Short	Long	Description
H/HT	Halt	Immediate STEP abort; enable off; cancel waits and remaining ; chain.
P	Pins	List PIN_*=GPIO lines (machine-readable, read-only).
\$/HL	Help	ASCII table of all commands ($\leq$ 80 columns).

Wire: one command per line (`\n`); chain with `;`; `#` comment to EOL (comment-only lines ignored); bare bool setters toggle; motion/settings silent on success; errors `!E:code message`.
Realtime (no newline): `? status · ! soft stop · Ctrl-X (0x18) soft reset`.
Status (`#X ...`): `I` idle · `A` accel · `M` cruise · `B` decel · `H` homing · `L` hard-limit · `D` disabled · `E` error. Moving: `#M/#A/#B pos speed accel [target]`.
Halt vs Stop: `MS/!` soft decel (enable kept); `H/HT` immediate abort, enable off, cancel waits.